

FOOD & GRAINS — INVENTORY MANAGEMENT SYSTEM

Software Requirements Specification | v1.0 | May 2026 | Prepared by: Prachi

OBJECTIVE

Build an inventory management system for food grains that handles:

- Barcode scan or Excel import for fast inward entry
- Auto-fetch of item details, expiry date & quantity on scan
- Automatic stock update on every inward and outward
- 3-level item structure: Category → Sub-category → Item
- Reports: item-wise, category-wise, date-wise

SYSTEM MODULES

Module	Key Function
Item Master	Category / Sub-cat / Item / Barcode / HSN / Unit
Barcode System	Scan → auto-fetch details, expiry, qty. Print labels
Purchase / Inward	PO → Receipt → Scan/Excel → Auto stock add
Sales / Outward	Dispatch → Scan → Auto stock deduct (FIFO)
Expiry Management	Batch-wise tracking, 30/60/90-day alerts
Reports & MIS	Daily, monthly, category-wise. Excel export
User Management	Role-based access: Admin, Purchase, Warehouse
Audit Trail	All entries logged with user + timestamp

ITEM CATEGORY STRUCTURE

Main Category	Sub Category	Items (Examples)
Food & Grains	Almonds	Mamro, Tucda, Running
	Rice	Basmati, Non-Basmati, Brown
	Pulses	Chana, Moong, Toor, Masoor, Urad
	Wheat & Flour	Wheat, Atta, Maida, Suji, Besan
	Dry Fruits	Cashew, Raisins, Walnuts, Pista
	Spices & Seeds	Whole Spices, Ground, Seeds
	Sugar & Jaggery	White Sugar, Jaggery, Khandsari

Item Fields: Item Code (auto), Barcode (auto), HSN Code, Unit, Variant/Grade, Purchase Rate, MRP, Min Stock Level

INWARD FLOW (Purchase Receipt)

	Step	Details
1	Raise PO	Vendor, items, qty, rate, delivery date
2	Receive Goods	Match against PO at warehouse
3	Invoice Entry	Invoice no., date, attach scanned copy
4A	Barcode Scan	Scan → auto-fetch item, batch, expiry, qty
4B	Excel Import	Fill template (Item Code, Barcode, Qty, Batch, Expiry) → upload
5	Auto Stock Update	System adds qty, records batch + expiry, posts entry
6	Confirm & Lock	Review summary → confirm → entry locked

OUTWARD FLOW (Sales Dispatch)

	Step	Details
1	Dispatch Order	Customer, items, qty required
2	Stock Check	System checks availability + selects FIFO batch
3	Barcode Scan	Scan item at dispatch → confirms correct batch
4	Auto Deduct	Stock deducted, outward entry posted automatically
5	Challan / Lock	Delivery challan generated → entry locked

REPORTS

- Daily Inward & Outward Report
- Current Stock Report (real-time)

DEVELOPMENT PHASES

Phase	Weeks	Deliverable
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TECHNICAL REQUIREMENTS

- Web-based (Chrome / Edge, desktop)

• Low Stock Alert Report • Expiry Alert Report (30/60/90 days) • Category-wise Stock Summary • Vendor Purchase History • Monthly MIS Dashboard • Audit Trail Report			• USB/Bluetooth barcode scanner — plug & play • Supports Code 128, EAN-13, QR codes • Excel import (.xlsx / .csv) with template • Excel export for all reports • Role-based login (Admin, Purchase, Warehouse, Sales, View-only) • Daily automated database backup • Audit trail: all actions logged with user + timestamp • Data retention minimum 7 years • Pet Pooja integration — Phase 3/4 (optional)
	1	1–4	
	2	5–8	
	3	9–12	
	4	13–16	
Coordinator: Prachi (requirement gathering, UAT, training)			